

Your Name:

ID #:

Worksheet: Limits at Discontinuity - Part 2

Note: for all the problems, identify the limit as a number, $-\infty$, ∞ or DNE (does not exist)

1. Show all the work to find the following limit.

(a) Find $\lim_{x \rightarrow 5} \frac{2x^2 - 7x - 15}{x - 5}$

(b) Find $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1}$

(c) Find $\lim_{x \rightarrow -2} \frac{x + 2}{\sqrt{x + 6} - 2}$

(d) Find $\lim_{x \rightarrow -1} \frac{\sqrt{x + 10} - 3}{x + 1}$

(e) $\lim_{x \rightarrow 0^+} \frac{3e^x}{\ln x}$

(f) Find $\lim_{x \rightarrow 0} \frac{x^2 + 8\sin(x)}{x}$

(g) Find $\lim_{x \rightarrow 0} \frac{\tan(6x^2) + \sin^2(2x)}{x^2}$

(h) Find $\lim_{y \rightarrow 0} \left(\frac{6}{y^2 + y} - \frac{6}{y} \right)$

(i) Find $\lim_{x \rightarrow 0} \frac{10\sin(2x)}{1 - \cos(2x)}$